Installation and User Guide

This guide provides step-by-step instructions for installing, configuring, and using the AI Sarthi Healthcare Assistant.

1. System Requirements

- **OS:** Windows 10/11, Linux, or macOS.
- **Python:** Python 3.9+ installed and accessible via PATH.
- **Hardware:**
  - **RAM:** Minimum 16 GB recommended.
  - **GPU:** An NVIDIA GPU with CUDA support is highly recommended for optimal model inference and essential for QLoRA fine-tuning.
- **Storage:** Sufficient disk space for ChromaDB vector storage and PyTorch model checkpoints.

2. Repository Setup

First, clone the repository to your local machine:

```
git clone https://github.com/vedant-bitbyte/healthcare-ai.git
cd healthcare-ai
```

3. Virtual Environment

It is highly recommended to use a virtual environment to isolate project dependencies.

Windows:

```
python -m venv venv
venv\Scripts\activate
```

Linux / macOS:

```
python3 -m venv venv
source venv/bin/activate
```

4. Dependency Installation

With the virtual environment activated, install the required packages using the provided requirements file:

```
pip install -r requirements.txt
```

Note: This installs Streamlit, PyTorch, LangChain, ChromaDB, Sentence Transformers, and evaluation libraries.

5. Install AI Sarthi as a Local Python Package

After installing the dependencies, install the project itself in editable mode:

```
pip install -e .
```

This step uses the project's pyproject.toml configuration to register the app and src modules as Python packages.

6. Environment Configuration

The system relies on environment variables for configuration.

1. Copy the provided example configuration file:cp .env.example .env
2. Open `.env` in a text editor. By default, it is pre-configured with:
    - `DEFAULT_MODEL=fine-tuned`
    - `HF_BASE_MODEL=microsoft/Phi-3-mini-4k-instruct`
    - `HF_ADAPTER_PATH=models/outputs`
    - `CHROMA_PERSIST_DIR=vector_store`

You typically do not need to alter these defaults unless you are changing directory structures or swapping the base model.

7. Model Setup

The system uses `microsoft/Phi-3-mini-4k-instruct` and overlays a custom LoRA adapter.

- Ensure that the pre-trained adapter weights are placed in the `models/outputs/` directory as specified by `HF_ADAPTER_PATH`.
- The application will automatically load the base model and apply the adapter weights during startup.

8. Dataset Setup

To add healthcare documents to the system's knowledge base:

1. Place your raw medical PDFs or CSV files into the `data/raw/` directory.

9. Document Ingestion

Process the raw documents into searchable text chunks by running the ingestion pipeline:

```
python -m src.ingestion.pipeline
```

This extracts the text and saves the processed chunks into `data/processed/`.

10. ChromaDB Setup

Build the local vector database so the RAG pipeline can retrieve the documents:

```
python -c "from src.vectorstore.chromadb_store import VectorStore;
VectorStore().index_chunks_directory()"
```

This creates or updates the ChromaDB instance located in the `vector_store/` directory.

11. Running Evaluation

To benchmark the model against the evaluation dataset:

```
python scripts/evaluate_model.py --model "fine-tuned"
```

The results will be logged to the console and saved to the `evaluation/` directory.

12. Running Streamlit

To launch the interactive web application, run:

```
streamlit run app/streamlit_app.py
```

A local web server will start, and the Streamlit UI will open in your default web browser (usually at `http://localhost:8501`).

13. Using the Chatbot

1. **Enter Query:** Type your healthcare policy or medical question into the chat input box at the bottom of the screen.

2. **Send Query:** Press Enter or click the send icon.

3. **Read Answer:** The AI Sarthi model will retrieve context and generate a response.

4. **View Sources:** Beneath the generated answer, the system will display the source documents utilized to formulate the response.

5. **Continue / New Chat:** You can ask follow-up questions in the same session, or refresh the page to start a new chat history.

14. Troubleshooting

- **Problem:** Application fails to start, citing missing models.
  - **Cause:** `adapter_config.json` or LoRA weights not found.
  - **Solution:** Verify that the trained adapter is correctly located in `models/outputs/` or that `HF_ADAPTER_PATH` in `.env` is accurate.
- **Problem:** Chatbot returns no context or empty sources.
  - **Cause:** ChromaDB is empty or unavailable.
  - **Solution:** Ensure you have placed documents in `data/raw/`, run the ingestion pipeline, and successfully built the ChromaDB store (Steps 8 & 9).
- **Problem:** `CUDA out of memory` during generation.
  - **Cause:** The system lacks sufficient GPU VRAM for the model.
  - **Solution:** Close other GPU-intensive applications. If running locally, you may need to utilize model quantization (e.g., bitsandbytes 4-bit) within the inference logic.

15. Project Maintenance

- **Adding New Documents:** Simply place new PDFs in `data/raw/`, rerun the ingestion pipeline, and execute the ChromaDB indexing script. The system will append the new vectors.
- **Updating the Model:** If you re-run the `train_phi3.py` script, the new weights will overwrite the old ones in `models/outputs/`. Restart Streamlit to load the new weights.

16. Healthcare Disclaimer

Disclaimer: AI Sarthi is an experimental AI Healthcare Assistant intended for informational and research purposes only. The generated outputs are based on the provided document contexts but may contain inaccuracies. This tool does **not** provide professional medical advice, diagnosis, or treatment. Always consult a qualified healthcare provider for medical decisions.